Temporal Referent Integrity: An Evaluation Diagnostic for Resolution Timing in Tool-Using Agents

Anonymous submission

Abstract

Tool-using agents often act on entities after the environment changes. We study instructions that either commit to a still-valid entity before a refresh or defer a selector until afterward. Evaluating only stable cases or only one instruction type can miss whether a controller handles both cases. We introduce *temporal referent integrity* (TRI), a diagnostic built from matched Preserve/Reevaluate pairs. Pair members share the states, selector, action, and transition but require opposite targets when the selector winner changes. Pair accuracy (PairAcc) counts a pair only when both targets are correct. In a post-primary controlled replication across three model backends on 240 authored tasks spanning ten schemas, an author-designed Generic controller substituted the refreshed winner in 41/66, 30/70, and 50/69 cases where both probes exposed the correct initial binding; a Compile-then-act probe with an explicit decision record made no substitutions in those cases. A SQLite tool-loop test traced the same error pattern into writes to the wrong entity. With equal calls and identical base actor inputs, exposing a composite decision block improved authored PairAcc for Qwen and GLM. Effects on pairs adapted from public benchmarks varied by model. Limited human agreement audits did not establish open-language transfer. TRI provides a controlled diagnostic for selective re-resolution; native prevalence and generalization to open-ended language remain unresolved.

Introduction

Tool-using agents often select an entity, observe an updated environment, and act only afterward. During this interval, an update can change which entity satisfies the original selector. We study the resulting distinction between instructions that commit to the earlier entity and instructions that defer selection until after the update.

Consider an email agent. If it selects the highest-priority unread email and then refreshes the mailbox, a later request to reply to “it” refers to the selected email. If the instruction instead refreshes first and selects afterward, the same description is evaluated on the refreshed mailbox. Suppose email A wins before refresh and email B afterward. The first instruction targets A and the second targets B, although the states, selector, transition, and action are otherwise identical.

The same distinction appears in file-management dialogue. After an assistant reports a results folder, “put the experiment data in this folder” remains bound to that folder even if staging folders appear. By contrast, “use the most recently created results folder” defers selection until after the filesystem update.

An agent trace may retain the initial entity identifier (ID), the refreshed state, and the action schema while leaving this distinction implicit. Reapplying the selector can replace a committed referent; retaining the old ID can violate an instruction that deferred selection. Stable-only and one-sided evaluations can leave this difference unidentified. A Preserve-only test can reward Always-Lock, a Reevaluate-only test can reward Always-Reevaluate, and stable winners make the two policies observationally equivalent.

We introduce temporal referent integrity (TRI) as a controlled matched diagnostic. TRI represents an action reference as either *bound* to an entity or *deferred* as a selector. A Preserve/Reevaluate pair holds the states, selector, action, and transition fixed while changing the instruction phrasing that encodes commitment timing. When the selector winner changes, the two members require opposite targets. PairAcc counts a pair only when both targets are correct. This construction tests selective re-resolution under a shared transition.

Deterministic policy extremes establish what the paired endpoint distinguishes. Conditional substitution then tests whether a controller replaces a correct initial binding with the refreshed winner. A post-primary 240-task controlled replication spanning ten schemas measures this behavior with three model backends. A separate SQLite study follows the selected ID through a mutation issued by the model.

Controller probes test whether an executable decision representation changes this behavior after the initial selection is visible. The primary comparison contrasts complete controller packages. An equal-call comparison then holds the base actor payload and call count fixed while exposing a composite decision block. PairAcc rises from 5/32 to 13/32 for Qwen and from 8/32 to 25/32 for GLM. Gains on pairs

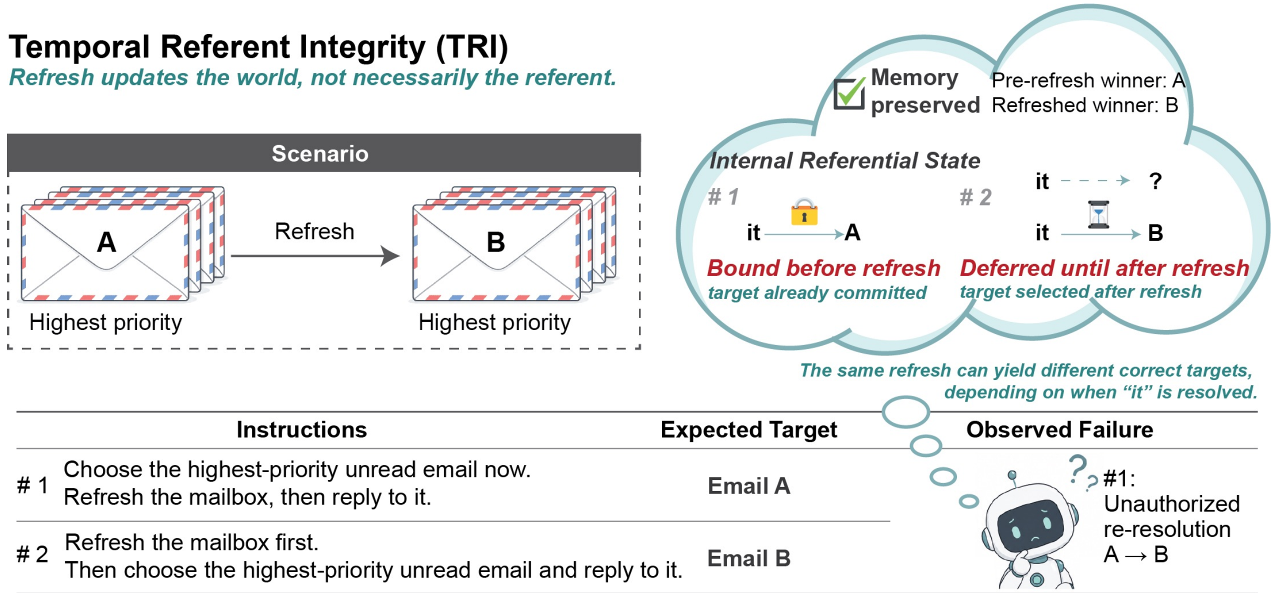

Figure 1: Temporal Referent Integrity under a shared transition. Email A wins before refresh and Email B afterward. Instruction 1 commits to A before refresh; Instruction 2 defers selection and targets B. Re-resolving Instruction 1 as B yields conditional target substitution despite the preserved state history.

adapted from public benchmarks are heterogeneous, and a strong event-order rule succeeds on the authored inventory but transfers poorly.

Our claim is diagnostic rather than architectural: the experiments test a controlled behavioral distinction, not the necessity of one record format, native-task prevalence, or open-language generalization.

The paper makes three contributions:

- We distinguish authorization from initial binding and post-commitment persistence, then identify selective resolution with crossed bound/deferred directions and opposite-gold PairAcc.
- A post-primary controlled replication links substitution after a correct binding to wrong-entity writes, while keeping initial binding and fallback errors outside the TRI endpoint.
- Equal-call probes estimate the effect of exposing one complete decision block; mixed transfer and a strong post-hoc rule bound claims about components, architecture, and scope.

Related Work

Reference, memory, and changing state. Discourse semantics separates a referent from a description whose denotation can change with context (Kamp and Reyle 1993; Grosz, Joshi, and Weinstein 1995). Coreference and entity-tracking work studies how systems recover and maintain these identity links (Lee et al. 2017; Joshi et al. 2020; Tang et al. 2026). Agent-memory systems retain facts and interaction traces (Packer et al. 2023; Sumers et al. 2024), while temporal-agent benchmarks test evidence refresh, delayed intentions, and belief updates (Cheng et al. 2026; Liu and Gabriel 2026; Ji et al. 2026; Xu et al. 2026). TRI asks a narrower question after memory and state are available: which representation is the action allowed to resolve at execution time?

Initial binding, persistence, and timing. Entity Binding studies initial tool-argument selection (Babu and Indukuri 2026b). Binding Drift assumes a primary referent has been committed, then tests persistence, reverification, and error propagation (Babu and Indukuri 2026a). TRI instead conditions on a correct initial binding, crosses bound and deferred authorization directions under the same transition, and jointly scores each opposite-gold Preserve/Reevaluate pair. It therefore evaluates whether re-resolution was authorized rather than proposing another persistence defense. Our Binding Drift author adaptation is neither an official reproduction nor an information-matched CTA baseline: it receives S_1 but neither S_0 nor the resolved old ID. Its complementary outcomes illustrate the interface boundary, not a performance advantage. Detailed counts are supplementary.

Behavioral and tool-agent evaluation. Behavioral tests and contrast sets expose capabilities that aggregate accuracy can miss (Ribeiro et al. 2020; Gardner et al. 2020). Stateful tool benchmarks add executable world changes, intermediate milestones, and target-level outcomes (Li et al. 2023; Liu et al. 2024; Lu et al. 2025; Trivedi et al. 2024; Yao et al. 2024; Sierra Research 2026; Yang et al. 2026). TRI changes resolution timing while holding the transition fixed. Joint scoring rejects two complementary policies, and the execution trace localizes errors after binding. Runtime ledgers and monitors (Uddin et al. 2026; Sohail and Haider 2026; Wang, Poskitt, and Sun 2025; Li et al. 2026) provide possible implementations. TRI evaluates the underlying resolution-time decision across such implementations.

Temporal Referent Integrity and Evaluation Identifiability

Let H be the instruction and interaction history, S_0 and S_1 the states before and after one refresh, r an action-target reference, and $q_r : S \to E \cup \{\perp\}$ its deterministic task resolver. Ties are excluded. We separate world evidence S, referential control state $C_H(r)$, and action validity $V_a(e, S)$. Immediately before refresh,

$$C_H(r) \in \{\mathrm{bound}(e_0), \mathrm{deferred}(q_r)\},\quad e_0 = q_r(S_0) \in E. \tag{1}$$

$\mathrm{bound}(e_0)$ records an identity commitment. $\mathrm{deferred}(q_r)$ records a selector to evaluate after refresh. A world update can change the selector winner without changing a bound referent:

$$q_r(S_0) \neq q_r(S_1) \;\not\Rightarrow\; \mathrm{bound}(e_0) \mapsto \mathrm{bound}(q_r(S_1)). \tag{2}$$

In the actionable core, where the required target exists and satisfies the action preconditions, the correct target is

$$T^* = \begin{cases} e_0, & C_H(r) = \mathrm{bound}(e_0) \wedge V_a(e_0, S_1), \\ q_r(S_1), & C_H(r) = \mathrm{deferred}(q_r). \end{cases} \tag{3}$$

Here $e_1 = q_r(S_1) \in E$ and the selected T^* satisfies $V_a(T^*, S_1)$; cases that violate these actionable conditions belong to the fallback-policy slice. Rows with $\neg V_a(e_0, S_1)$ form a separate fallback-policy slice. Reject, Clarify, and licensed reselection are execution choices beyond the referential identity estimand.

Observable substitution. We count conditional substitution only on Preserve rows with four observable properties: the initial target is correct; refresh completes; the selector winner changes; and the old target remains present and action-valid. A substitution occurs when the final target equals the refreshed winner. This endpoint separates post-binding replacement from initial grounding and validity errors. A state diff then tests whether the selected target receives the write.

Restricted identifiability observation. Within the exact-target policy class {Always-Lock, Always-Reevaluate, Selective}, evaluation requires at least one changed-winner Preserve row and one changed-winner Reevaluate row to distinguish Selective from both extremes. A two-row set is sufficient and minimal. Matching the rows on S_0, S_1, q, and a additionally controls task and transition content; it is not needed for the cardinality claim. The observation assumes deterministic policies, and the short argument is supplementary.

Stable tasks make lock and reevaluation observationally equivalent. A one-sided changed-winner test can reward one unconditional extreme. The crossed matched regime breaks both equivalences. For pair i, let Y_{Pi} and Y_{Ri} indicate exact correctness on its Preserve and Reevaluate members. Then

$$\mathrm{PairAcc} = \frac{1}{N}\sum_i Y_{Pi}Y_{Ri},\quad \max(0, A_P + A_R - 1) \leq \mathrm{PairAcc},\quad \mathrm{PairAcc} \leq \min(A_P, A_R), \tag{4}$$

Evaluation regime	Lock	Reevaluate	Identifies?
Stable only	can pass	can pass	No
Changed Preserve only	can pass	fails	No
Changed Reevaluate only	fails	can pass	No
Matched changed PairAcc	zero	zero	Yes

Table 1: Policy identifiability. Stable or one-sided evaluation can favor unconditional policies; matched changed-winner PairAcc rejects both extremes within the three-policy class.

where $A_P = N^{-1}\sum_i Y_{Pi}$ and $A_R = N^{-1}\sum_i Y_{Ri}$ use the same N complete pairs. The marginals alone do not determine joint correctness. Crossed Preserve/Reevaluate coverage supplies the policy discrimination; matching controls task content, and PairAcc summarizes co-correctness within each pair. Table 1 gives the restricted policy comparison.

Aggregate certification gap. Suppose an N-row evaluation contains n disjoint complete actionable changed pairs and has aggregate end-to-end (E2E) accuracy A. Counting failed pairs gives the sharp certificate $\mathrm{PairAcc} \geq \max\{0, 1 - N(1-A)/n\}$. Hence $A = 1 - n/N$ remains compatible with zero PairAcc: one error in every critical pair can be diluted by non-pair rows. This is a worst-case guarantee, not evidence that aggregate selection fails in every candidate set.

Execution-risk decomposition. Let O denote a strict changed-winner opportunity, B a correct observable initial binding, U substitution to the refreshed winner, and X execution of the resulting unauthorized write. The chain rule gives

$$\Pr(O, B, U, X) = \Pr(O)\Pr(B \mid O) \times \Pr(U \mid O, B)\Pr(X \mid O, B, U). \tag{5}$$

Conditional substitution estimates the third factor in the tested controller; it does not estimate opportunity prevalence, execution realization, or total tool-agent risk. Complete arguments and assumptions are supplementary.

We separately report exact target/final state, wrong-target writes, invalid attempts, and unnecessary rejection; always rejecting avoids writes but fails every actionable task.

Methods

Diagnostic Construction

Each domain specification defines an entity table, a natural-language selector, its ranking or Boolean criterion, an action, and explicit action preconditions. The generator first evaluates the selector on S_0 to obtain e_0, applies a controlled transition, and evaluates it on S_1 to obtain e_1. A language template determines whether the action target is e_0, e_1, or $\perp$. Paired tasks share the domain, both states, selector, action, and schema. Their instruction phrasing differs to encode whether the target is committed before refresh or selected afterward.

The crossed core combines Preserve/Reevaluate with an unchanged or changed selector winner. Stable controls expose unnecessary reactions to refresh. Changed-winner and name-collision cases create opposite-target pairs while retaining valid stable IDs. The Matched Timing Diagnostic contains 128 actionable rows and 32 Remove/Invalidate rows. The latter form a separate fallback-policy slice. The Cross-Schema Controlled Replication contains 40 state clusters across ten schemas. Each cluster contributes six rows: two reference modes crossed with Stable, Flip, and name-collision transitions. Flip and name-collision each form one changed-winner pair per cluster, yielding 80 changed pairs and the PairAcc denominator of 80 in Figure 4. Complete templates and transition taxonomies are supplementary.

Figure 2 summarizes the matched construction, shared probe interface, and distinct observable readouts; it is an evaluation workflow, not a controller architecture.

Measurements and Denominators

The frozen primary estimand is the Qwen all-row end-to-end (E2E) package difference between Generic and Lifecycle-Gated. PairAcc, actionable exact-target accuracy, and conditional substitution are the main diagnostic endpoints; initial binding, rejection, invalid attempts, and wrong-target writes remain separate. Results use intention-to-treat scoring and cluster-bootstrap intervals by state or workflow. Later intervals are descriptive. PairAcc measures co-correctness within frozen pairs, so it is not determined by the two marginals alone.

A deterministic executor checks whether each selected target becomes the written target in the refreshed state. A separate 40-task SQLite tool loop records model-driven final states, wrong writes, invalid attempts, and collateral changes.

Controller Probes

Generic Structured Ledger (Generic) retains the instruction, selected ID, entity snapshot, selector, action, and preconditions. It does not encode resolution mode. Compile-then-act (CTA) emits the binding time, selector, and bound ID before refresh. Lifecycle records add validity and fallback fields, while actor-mediated and gated variants test soft and hard use of that record. These large language model (LLM) controllers are operational probes for the same timing decision. The typed record and gate branches are supplementary. Generic exposes both the old selected entity and the current selector result but does not prescribe a timing policy.

Always-Lock and Always-Reevaluate provide deterministic policy extremes. The former retains the S_0 target when valid; the latter applies q to S_1. A timing reminder tests whether ordinary history can expose the distinction without a structured record. Rule* is a deterministic event-order parser written after inspection of earlier errors and is reported as post-hoc.

Controllers receive the instruction's natural-language selector. Gold targets, normalized selector fields, and pre- and post-refresh winner IDs are withheld. The primary Qwen comparison is a package contrast with unequal calls because Lifecycle-Gated skips an actor call on valid Preserve rows. Later equal-call experiments compare History-only and Decision-visible actors using the same base payloads, states, tool schemas, and call count. The visible block jointly contains predicted mode, bound ID, and selector restatement;

Condition	Equal-call interface per task
Shared calls	One compiler call and two actor calls; actor order alternates by task index.
Actor base	Byte-identical instruction, states, selected ID, selector, action, and tool schema.
History-only	Base payload only.
Decision-visible	Same base plus parsed `reference_mode`, `bound_target_id`, and `selector`.
Decision-enforced	Zero-call deterministic transform, not a third actor condition.

Table 2: Equal-call actor interface. The contrast estimates the conditional effect of exposing the complete decision block; it does not identify any field or framing component separately.

Decision-enforced applies it offline. The same interface is used for authored, rewritten, and public-benchmark-adapted tasks. Prompts, payloads, and exact scoring rules are supplementary. The PDF supplement gives the verbatim interface; prompts, payloads, raw outputs, and scoring code are in the anonymous Code and Data Supplement.

Construct and transfer protocols. One adult volunteer rewrote 50 sampled scalar instructions. Three other adults independently labeled 100 randomized original/rewrite items using opaque IDs without condition metadata or gold. Referent identity is analyzed separately from Reject, Clarify, or reselection. Participants gave informed consent; released responses are de-identified. No formal institutional review or exemption determination was obtained. The transfer audit uses 30 author-constructed pairs over states and tool schemas from three public resources, with matched interfaces and call counts.

Evidence Status

The Qwen package comparison is primary/frozen, with a later GLM replication. The ten-schema attribution, equal-call contrasts, human agreement audits, and external audits are post-primary; Rule* is post-hoc. The supplement gives the complete chronology, denominators, intervals, and failure accounting. Qwen3.5-122B-A10B (Qwen), GLM-5.1 (GLM), and, in later transfer tests, DeepSeek-V4-Pro (DeepSeek) use one temperature-zero protocol. The artifact contains frozen inventories, prompts, outputs, reports, hashes, and the error taxonomy.

These evidence families answer different questions and are not pooled. Deterministic controls establish policy discrimination; authored runs spanning ten schemas locate substitution after correct binding; the SQLite subset tests execution; equal-call contrasts test the composite decision block; and external adaptations and audits bound transfer and occurrence.

Results

The results ask whether the paired design identifies selective behavior, whether substitution reaches execution, whether decision visibility changes equal-call outcomes, and how far the construct transfers.

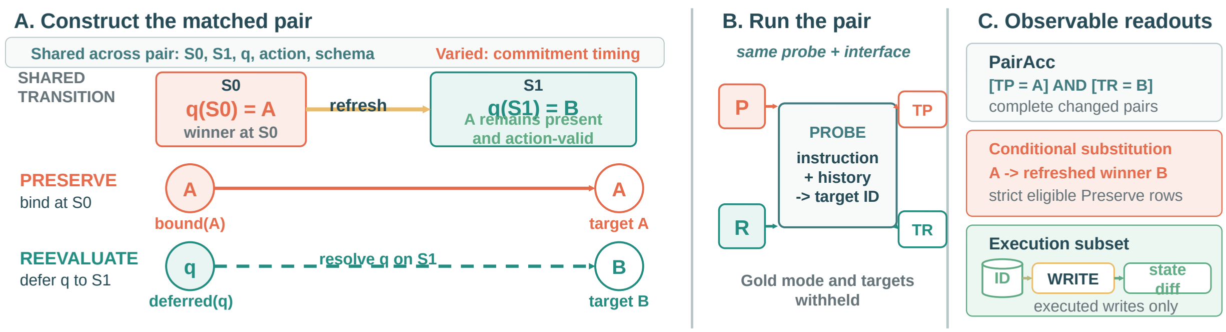

Figure 2: TRI diagnostic construction and observable endpoints. A changed-winner pair fixes S_0, S_1, selector q, action, schema, and controller interface. A wins in S_0; B wins in S_1, while A remains present and action-valid. Preserve binds A before refresh, whereas Reevaluate defers q and targets B afterward. PairAcc requires both outputs to be correct, conditional substitution isolates $A \to B$ after a correct binding, and the execution subset records which entity receives the write. Gold resolution modes and targets are withheld from the probe.

Claim	Direct evidence	Status	Boundary
Identify selective resolution	Changed-winner pairs with opposite targets and deterministic extremes	Formal + offline audit	Exact targets within three policy extremes; no internal-mechanism identification
Locate substitution and writes	Ten-schema audit and SQLite tool-loop trajectories	Post-primary + secondary frozen	Author-designed controller and tasks; not a prevalence estimate
Test an executable decision block	Equal-call History-only vs. Decision-visible conditions	Post-primary, frozen before own calls	Initial ID already visible; block and framing are not separated
Bound transfer and practical scope	Human rewrites, public-benchmark adaptations, low-intervention tests, and public benchmark audits	Post-primary/descriptive	Mixed transfer, failed human follow-up, and uncalibrated native recall

Table 3: Claim-to-evidence map. Evidence families answer distinct questions and retain their chronology; no row upgrades another row's evidentiary status.

RQ1: Does Pairing Identify Selective Resolution?

Policy discrimination. Always-Lock and Always-Reevaluate each score 60.0% in aggregate, but succeed on complementary instruction types and score 0/32 changed-winner PairAcc (Figure 3). Across five candidate sets, aggregate E2E and PairAcc nevertheless choose the same tested policy; there is no ranking reversal. Rule* is post-hoc, and its strong score limits algorithmic novelty.

RQ2: Does Substitution Reach Execution?

Primary package comparison. The primary comparison changes the complete controller package and uses different numbers of model calls. Across 160 tasks, Qwen E2E changes from 103/160 for Generic to 157/160 for Lifecycle-Gated (+33.8 points; language-template-cluster 95% CI [18.1, 49.4]); the later GLM replication changes from 115/160 to 160/160 (+28.1 points; 95% CI [18.1, 38.1]). On 128 actionable tasks, Lifecycle-Gated and CTA reach 125–128/128, versus 95/128 and 93/128 for Generic. Fallback cases are supplementary. This is a package contrast, not the effect of one timing field.

Conditional target substitution. We condition on a correct initial ID, completed refresh, distinct new winner, and surviving valid old target, excluding initial-binding, tool-order, and validity failures. In the 240-task, ten-schema replication, Generic substitutes on 41/66, 30/70, and 50/69 rows eligible under both controllers; CTA substitutes on none (Figure 4). The pattern appears for Qwen, GLM, and DeepSeek under the same author-designed controller.

Executed target consequences. In the 40-task SQLite tool-loop test, Generic reaches 27/40 correct final states with 13 wrong-target writes for Qwen, and 26/40 with eight for GLM. Among strict opportunities, refreshed-winner writes occur in 8/8 and 6/8 Changed cases, versus 0/4 Stable controls for each model (Figure 5). Remaining wrong writes are confined to fallback cases; GLM also produces six unneeded rejections. These trajectories locate the failure between binding and action.

A fixed-executor replay turns every Generic substitution in the 240-task replication into a wrong-target write. This target-to-write consistency check is supplementary.

RQ3: Does Decision Visibility Change Outcomes under Equal Calls?

The equal-call test holds base payloads, states, and tool schemas fixed. Exposing predicted mode, bound ID, and selector restatement improves authored PairAcc and actionable

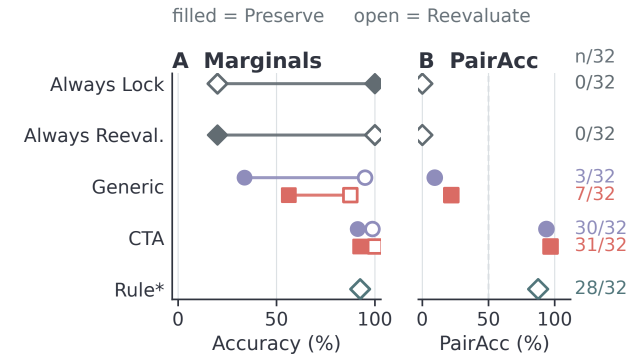

Figure 3: Marginal accuracy can mask incorrect resolution policies. Panel A reports Preserve/Reevaluate marginals (80 rows each); Panel B reports both-correct counts for 32 actionable changed-winner pairs. Shape denotes Qwen, GLM, or deterministic policies; Rule* is post-hoc.

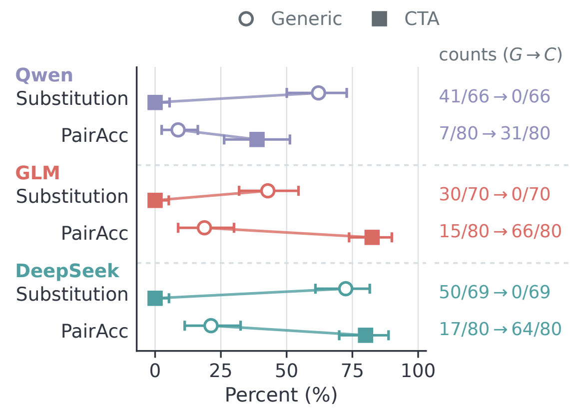

Figure 4: Ten-schema conditional outcomes after correct initial binding. Within each model, connected markers compare Generic (open circles) with CTA (filled squares): substitution on rows eligible under both controllers (Wilson 95% intervals) and changed-winner PairAcc over 80 complete pairs (cluster-bootstrap 95% intervals). Right labels give Generic→CTA counts with endpoint-specific denominators. CTA has no common-eligible substitutions; these are not safety rates.

E2E for both models (Figure 6), but estimates the complete block rather than any field. Hard enforcement repairs and harms some Qwen rows; this is a descriptive comparison, not a mechanism-separating test. The full decomposition is supplementary.

Across all matched-call audits, both actors receive the same initial ID and the visible selector restates the base selector. Compiler-output strata are supplementary and descriptive.

RQ4: Does the Finding Transfer?

Construct scope. In the earlier three-annotator convenience sample, majority–gold agreement is 98.0% on 50 dynamic items and 86.7% on 30 anchored actionable items, but 55.0% on 20 action-invalid REJECT items. A frozen follow-up missed its eligibility gate and produced no predeclared item endpoint; its 38.6% retained-label agreement is descriptive. We therefore center actionable referent identity, treat fallback policy separately, and do not take this limited agreement as evidence of open-language transfer.

Human rewrites. On the 48 rewrites with determinate majority labels, CTA reaches 89.6% for Qwen and 93.8% for GLM. Equal-call actionable E2E is 30/40 for both Qwen conditions and rises from 31/40 to 39/40 for GLM. Only three changed pairs are complete, so PairAcc is descriptive; one volunteer does not establish open-language transfer.

Pairs adapted from public benchmarks. The 30-pair test uses states and tool schemas from STATE-Bench, AgentDojo, and ToolSandbox; the authors constructed the interventions. Effects are heterogeneous (Figure 6): Qwen is null, DeepSeek intervals cross zero, and only GLM has an actionable-E2E interval excluding zero. These adaptations are neither native tasks nor prevalence evidence.

Representation and rule boundaries. CTA, Lifecycle, and a full-history timing reminder improve some authored conditions, but no representation dominates across models and datasets. Rule*, written after error inspection, reaches 92.5% on Matched Timing Diagnostic, 96.0% on rewrites, and 91.7% on Cross-Schema Controlled Replication, showing usable event-order cues in the authored inventory. After freezing, it reaches only 15/60 exact targets and 2/30 PairAcc on the public-benchmark adaptations, where unfamiliar rankings introduce parsing and grounding errors.

A post-primary oracle decomposition identifies a separate grounding boundary. On an 80-row unseen-schema stress set, Qwen mode is 80/80 but the Preserve bound ID is 21/40. Replacing the ID raises gated accuracy from 66/80 to 78/80; replacing mode does not. In this Qwen 80-row stress set, the remaining loss is concentrated in selector grounding.

Coverage in native tasks. An extension finds zero substitutions in four conditions over 64–87 eligible opportunities. Across 20 STATE-Bench and AgentDojo clusters (Microsoft Research 2026; Debenedetti et al. 2024), only Qwen ordinary history on AgentDojo shows substitutions: 2/7 changed rows versus 0/7 matched Stable rows. The other combinations of repository and model are zero, and no method consistently improves execution accuracy. These are conditional endpoints, not overall safety rates.

An author-built retrieval audit finds no case meeting all eligibility conditions in six public benchmarks. Synthetic controls verify the checklist, but natural retrieval recall is uncalibrated.

Composition and repeatability. Composition tests are mixed: a two-refresh Qwen test favors Generic over scalar Lifecycle (32/40 vs. 28/40), while held-out role indexing improves 35/40 to 39/40. Across three temperature-zero passes, CTA remains above Generic and records no substitutions. Full results are supplementary.

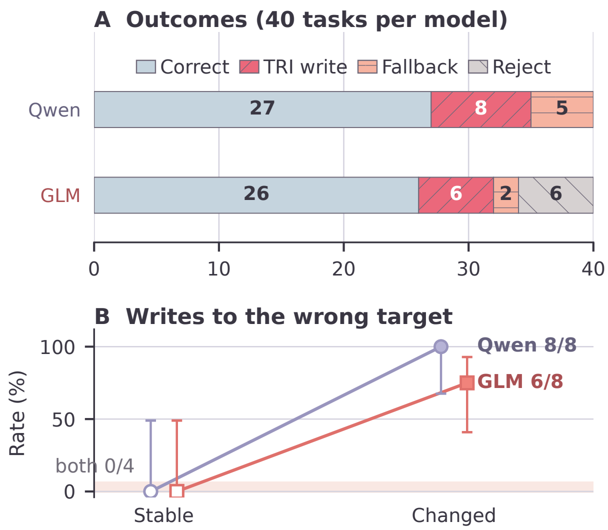

Figure 5: Secondary/frozen 40-task SQLite outcomes for Generic. Panel A partitions trajectories into correct states, TRI wrong-target writes, fallback writes, and rejections. Panel B gives strict-case write rates with Wilson 95% intervals: Stable is 0/4 for both models; Changed is 8/8 for Qwen and 6/8 for GLM. Model-issued tool calls are not a prevalence estimate.

Discussion

TRI pairs instructions that share a transition but place the action reference on opposite sides of it. PairAcc tests joint correctness, conditional substitution localizes post-binding failures, and SQLite traces identify the written entity. In these candidate sets, the diagnostic reveals complementary success patterns without changing the controller chosen by aggregate E2E; it need not reverse every ranking.

The composite decision block improves authored PairAcc, but the probes do not identify a unique architecture or field-specific effect. Enforcement can preserve a wrong compiled decision. Rule*'s authored strength and external failure are consistent with a boundary between event-order extraction and selector grounding.

Transfer remains model-dependent: only GLM shows a clear external-adaptation E2E gain, the low-intervention extension is null, and the retrieval audit finds no native eligible case. Prevalence therefore remains unresolved.

Limitations

The diagnostic covers single-refresh scalar tasks and simplifies ambiguity, repair, partial observability, identity migration, and composition. The ten-schema replication retains authored templates; volunteer rewrites contain only three complete changed pairs, and a model-authored audit yields none accepted by both judges. Open-language generalization remains unresolved.

The results-folder example is illustrative and was selected from an author–assistant workflow; it is not counted as independent natural-log or prevalence evidence.

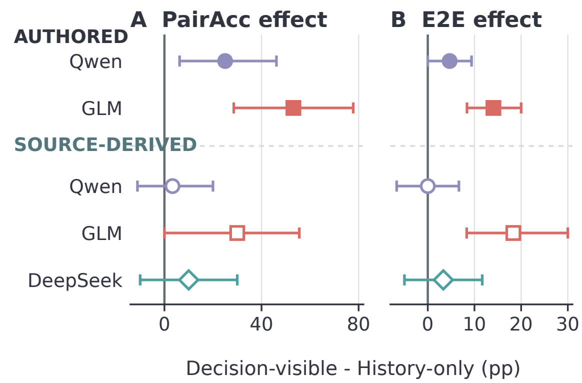

Figure 6: Equal-call effect of exposing the complete decision block. Panels report changed-winner PairAcc and actionable E2E differences with cluster-bootstrap 95% CIs. Filled/open marks denote authored/source-derived tasks; shape denotes model. Effects are unpooled, and adaptations are not native tasks or prevalence evidence.

Agreement is higher for referent identity than for fallback policy; Reject, Clarify, and licensed reselection may each be reasonable when the old target is action-invalid. Multiple refreshes and referent roles require richer state, and the scalar record does not compose automatically.

External nulls may reflect low prevalence, coverage, or retrieval misses; recall is uncalibrated. The primary package comparison uses unequal calls. The equal-call intervention is composite and begins after supplying the initial ID, while adaptations retain author-supplied timing contrasts. Provider weights lack immutable revision identifiers, and the repeat subset is too sparse for repeated PairAcc.

Conclusion

TRI pairs bound and deferred references under the same transition to make selective re-resolution measurable. PairAcc distinguishes selective behavior from unconditional locking and reevaluation, while conditional substitution and SQLite traces connect target replacement to writes on the wrong entity. In post-primary equal-call tests, exposing the complete decision block improves authored PairAcc for Qwen and GLM, while transfer to public-benchmark adaptations varies by model. Evidence is limited to the tested single-refresh scalar policies and complete-block intervention. It does not establish native prevalence, open-language transfer, or the necessity of a unique architecture.

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